

from where it was created by tectonic movement. See also *island arc*, *subduction*.

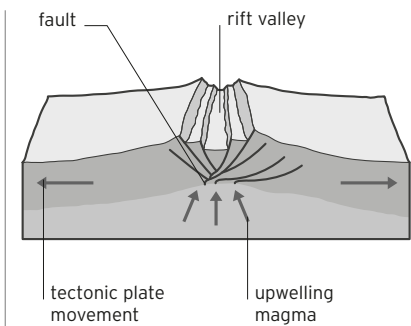
**RESURGENCE** The place where an underground watercourse emerges to become a surface stream or river. See also *karst*, *limestone*.

**REVERSE FAULT** An angled fault where one side has been pushed up relative to the other side. Reverse faults occur where two

blocks of rock have been forced together by compression. See also *fault* (panel).

**RIA** A former river valley that has been drowned by a rise in sea level to form an inlet of the sea.

**RIDGE OF HIGH PRESSURE** A wedge-shaped area of relatively high atmospheric pressure, an extension of an anticyclone. See also *anticyclone*.



**RIFT VALLEY** ▲ A large block of land that has dropped vertically relative to the regions on either

side. A rift valley, or graben, forms as the result of horizontal extension of Earth's crust and normal faulting. See also *normal fault*.

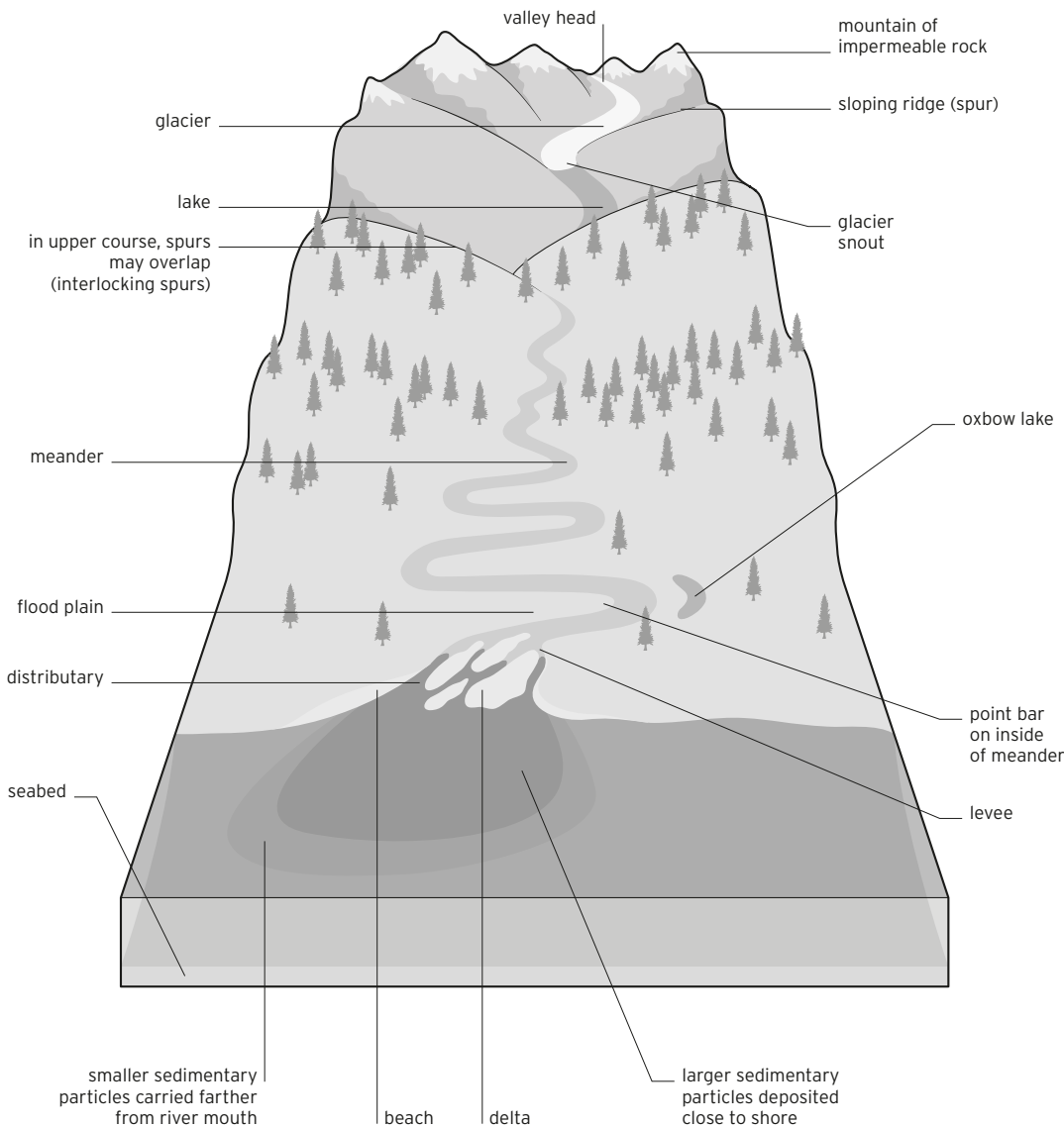
**RIVER BASIN** A region drained by a river and its tributaries where sediment accumulates.

**RIVER VALLEY** See panel, left.

**ROCK** A natural material made up of one or more minerals. See also *mineral*.

RIVER VALLEY

A river valley is a depressed area, longer than it is wide, that has been carved by a river. A valley's shape depends on terrain, geology, and whether it has been modified by ice. Since a river has an upper, middle, and lower course, the shape of its valley reflects this. The upper section is V-shaped unless glacial ice has carved it into a U shape. Slopes are usually less steep in the middle course, and often very gentle in a valley's lower course, where the river may meander and deposit sediment on a floodplain, often dividing into channels called distributaries to form a delta. A river flows out to sea through its mouth and deposits more sediment on the seafloor.



S

**SALINITY** The concentration of dissolved salts in water, usually expressed in parts per thousand.

**SALT MARSH** See panel, right.

**SANDSTONE** A sedimentary rock composed mostly of sand-sized grains of quartz or feldspar that have been cemented together.

**SASTRUGI** Sharp, irregular grooves or ridges on an ice or snow surface. Sastrugi, or zastrugi, patterns are formed by wind erosion, particularly the blasting of the surface by ice crystals.

**SAVANNA** A mixed grassland-woodland ecosystem of the tropics where the trees are not close enough to form a closed canopy. Often found between deserts and tropical forests, savannas are characterized by seasonal rainfall and regular wildfires. See also *cerrado*.

**SEA ARCH** A natural arch formed in a sea cliff by marine erosion.

**SEA CAVE** A cave eroded into a sea cliff by marine erosion.